CSE400 Lecture-20 Scribe

L20.pdf
PDF

Generate a lecture scribe intended to serve as exam-oriented reference material for CSE 400: Fundamentals of Probability in Computing, Lecture 20.

Use ONLY the provided context (Lecture 20 PPT). Do not use any CP questions or any material not explicitly present in the lecture slides or the relevant textbook sections covered in this lecture.

The scribe must strictly follow the lecture's structure, scope, notation, slide order, and sequence exactly as presented in the PPT. Prioritize correctness and faithfulness to the lecture content. Do not invent definitions, results, proofs, examples, explanations, intuition, or interpretations that are not explicitly present in the slides.

While writing the scribe, reason step by step through all definitions, assumptions, theorem statements, proofs, derivations, and worked examples exactly as shown in the lecture slides. Make every logical and algebraic step explicit if it appears in the slides, so the document can be reliably used for closed-notes exam revision.

Include:
- All slide titles and section headings in order
- All definitions and notation introduced
- All stated assumptions and conditions
- All theorems or results exactly as written
- Step-by-step proofs or derivations where they appear
- Worked examples with all intermediate steps, only if they are part of the lecture

Do not simplify, reinterpret, summarize, compress, or add explanations beyond what is explicitly covered in the slides. Do not introduce new examples or alternative solution methods.

Organize the scribe clearly using structured section headings corresponding to the lecture slides so that a student studying only this document can accurately revise Lecture 20 for an exam.

Below is a **lecture scribe for CSE400 – Lecture 20: Linear Transformations and Expectations of Multiple Random Variables**, written **strictly from the provided lecture slides (L20.pdf)** and **following the slide structure, notation, and sequence exactly**.

CSE 400: Fundamentals of Probability in Computing

Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

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Outline

- Statistical Representation of Random Vectors
- Mean Vector, Correlation, and Covariance Matrices
- Linear Transformations
- Mapping properties under $Y = AX + b$
- Multivariate Gaussian Distributions
- From 1D/2D cases to N-dimensional Joint PDFs
- Special Cases and Properties
- Implications of zero correlation and matrix factorization

Random Vector Representation

Vector Form of Multiple Random Variables

For random variables

$$X_1, X_2, ..., X_N$$

we represent them as a **vector**

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Definition

- X is called a **random vector**.

Properties

- It provides a **compact representation of multiple random variables**.

Mean Vector

Expectation of a Random Vector

Mean Vector Definition

The **mean vector** of X is

$$\mu_X = E[X]$$

Interpretation

- Represents the **average value of the random vector**.

- Each element corresponds to the **mean of one random variable**.

Example Structure

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

Correlation Matrix

Correlation Between Multiple Random Variables

Correlation Matrix Definition

$$R_{XX} = E[XX^T]$$

where

$$i, j \leq N$$

and

$$R_{XX}(i,j) = E[X_i X_j]$$

Matrix Representation

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \dots & E[X_1X_N] \\ E[X_2X_1] & E[X_2X_2] & \dots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \dots & E[X_NX_N] \end{bmatrix}$$

Property

- R_{XX} is a **symmetric matrix**.

Understanding the Covariance Matrix

The Relationship Map of Random Vectors

Mathematical Definition

For a random vector X, the **covariance matrix** is

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

Matrix Form

$$C_{XX} = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) \\ Cov(X_2, X_1) & Var(X_2) \end{bmatrix}$$

Property

Symmetry property:

$$Cov(X_i, X_j) = Cov(X_j, X_i)$$

Thus the matrix is a **mirror of itself** across the diagonal.

Linear Transformation of Random Vectors

Linear Model

Consider the transformation

$$Y = AX + b$$

Components

- A : Transformation matrix
  - Scales and rotates the data
- b : Constant vector
  - Shifts/translates the center
- X : Input random vector

Element-wise Representation

For N-dimensional vectors:

$$Y_1 = a_{11}X_1 + a_{12}X_2 + \cdots + a_{1N}X_N + b_1$$

$$Y_2 = a_{21}X_1 + a_{22}X_2 + \cdots + a_{2N}X_N + b_2$$

$$Y_N = a_{N1}X_1 + a_{N2}X_2 + \cdots + a_{NN}X_N + b_N$$

Mean of Transformed Vector

Mean Transformation

For the linear transformation

$$Y = AX + b$$

the mean vector is

$$\mu_Y = E[Y]$$

Derivation

Substitute $Y = AX + b$

$$\mu_Y = E[AX + b]$$

Apply linearity of expectation:

$$E[AX] + E[b]$$

Since b is constant:

$$E[b] = b$$

Thus,

$$\mu_Y = A\mu_X + b$$

Result

$$\boxed{\mu_Y = A\mu_X + b}$$

Correlation Matrix Transformation

Definition

$$R_{YY} = E[YY^T]$$

Substitute $Y = AX + b$:

$$R_{YY} = E[(AX + b)(AX + b)^T]$$

Expansion of Terms

$$R_{YY} = E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

Applying Expectation

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^TA^T + bb^T$$

Final Result

$$\boxed{R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T}$$

Covariance Matrix Transformation

Key Observation

The deviation from the mean is independent of the shift b:

$$Y - \mu_Y = (AX + b) - (A\mu_X + b)$$

Thus

$$Y - \mu_Y = A(X - \mu_X)$$

Derivation

Covariance definition:

$$C_{YY} = E[(Y - \mu_Y)(Y - \mu_Y)^T]$$

Substitute:

$$C_{YY} = E[A(X - \mu_X)(X - \mu_X)^T A^T]$$

Simplified Result

$$\boxed{C_{YY} = AC_{XX}A^T}$$

Important Observation

- The **shift vector** b does **not affect the covariance**.

Summary: Effects of Linear Transformation

If

$$Y = AX + b$$

then

Property	Result
Mean	$\mu_Y = A\mu_X + b$
Covariance	$C_{YY} = AC_{XX}A^T$
Correlation	$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$

Example: Numerical Linear Transformation

Given

$$Y = AX$$

Compute

- Mean vector of Y
- Correlation matrix of Y
- Covariance matrix of Y

Solution (a): Mean Vector

Using

$$\mu_Y = A\mu_X$$

$$\mu_Y = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Compute element-wise:

First component

$$(1)(2) + (0)(-1) + (-1)(1)$$

Second component

$$(0)(2) + (-1)(-1) + (1)(1)$$

Third component

$$(-1)(2) + (1)(-1) + (0)(1)$$

Thus

$$\mu_Y = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

Solution (b): Correlation Matrix

Using

$$R_{YY} = AR_{XX}A^T$$

Step 1: Compute AR_{XX}

$$AR_{XX} = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -1 & 8 & 0 \end{bmatrix}$$

Step 2: Multiply by A^T

$$R_{YY} = \begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix}$$

Solution (c): Covariance Matrix

Using

$$C_{YY} = R_{YY} - \mu_Y \mu_Y^T$$

Outer Product of Mean

$$\mu_Y \mu_Y^T = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

Final Subtraction

$$C_{YY} = \begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$$

Result:

$$C_{YY} = \begin{bmatrix} 16 & -8 & -6 \\ -8 & 10 & -2 \\ -6 & -2 & 10 \end{bmatrix}$$

Homework Assignment: Finding the Mean

Given

$$C_{XX} = \begin{bmatrix} 5 & -1 & 2 \\ -1 & 5 & -2 \\ 2 & -2 & 8 \end{bmatrix}$$

$$R_{XX} = \begin{bmatrix} 6 & 1 & 1 \\ 1 & 9 & -4 \\ 1 & -4 & 9 \end{bmatrix}$$

Task

Find the mean vector μ_X.

Hint

Use the relation

$$C_{XX} = R_{XX} - \mu_X \mu_X^T$$

Rearrange:

$$\mu_X \mu_X^T = R_{XX} - C_{XX}$$

Identify the elements of μ_X from the **diagonal of the resulting matrix**.

Gaussian Distributions: 1D and 2D

1. Univariate Gaussian (1D)

For a random variable X:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

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2. Joint Gaussian (2D)

For two random variables X and Y:

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho_{xy}^2}} \exp\left(-\frac{1}{2(1-\rho_{xy}^2)}\left[\frac{(x-\mu_x)^2}{\sigma_x^2} - \frac{2\rho_{xy}(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} + \frac{(y}{}\right.\right.$$

Multivariate Gaussian Distribution

For an N-dimensional random vector

$$X = [X_1, X_2, \ldots, X_N]^T$$

General Vector PDF

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^N \det(C_{XX})}} \exp\left(-\frac{1}{2}(x-\mu_X)^T C_{XX}^{-1}(x-\mu_X)\right)$$

Components

- μ_X : Mean vector $(N \times 1)$
- C_{XX} : Covariance matrix $(N \times N)$

Example: 2D Joint Gaussian PDF

Given

Mean vector

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

Covariance matrix

$$C_{XX} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

Step 1: Determinant

$$\begin{aligned} \det(C_{XX}) &= \sigma_1^2\sigma_2^2 - (\rho\sigma_1\sigma_2)^2 \\ &= \sigma_1^2\sigma_2^2(1-\rho^2) \end{aligned}$$

Step 2: Inverse of Covariance Matrix

$$C_{XX}^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1/\sigma_1^2 & -\rho/(\sigma_1\sigma_2) \\ -\rho/(\sigma_1\sigma_2) & 1/\sigma_2^2 \end{bmatrix}$$

Step 3: Quadratic Form

$$Q(x) = (x-\mu)^T C_{XX}^{-1}(x-\mu)$$

After algebraic simplification:

$$Q(x) = \frac{1}{1-\rho^2}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]$$

Substituting into the multivariate Gaussian formula yields the **bivariate Gaussian PDF**.

Example: Uncorrelated Gaussian Random Variables

Statement

For a vector of **jointly Gaussian random variables** that are **mutually uncorrelated**, prove that the joint Gaussian PDF becomes

$$f_X(x) = \prod_{n=1}^{N} f_{X_n}(x_n)$$

Interpretation

- If variables are **uncorrelated**:

$$Cov(X_i, X_j) = 0 \quad \forall i \neq j$$

Covariance Matrix Form

The covariance matrix becomes **diagonal**:

$$C_{XX} = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_N^2 \end{bmatrix}$$

Result

Substituting the diagonal covariance matrix into the multivariate Gaussian PDF causes the exponent to **separate into a sum of independent terms**, yielding

$$f_X(x) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$

Thus the **joint distribution factorizes into the product of individual Gaussian PDFs**.

Homework

Prove:

$$Var[X + Y] = Var[X] + Var[Y] + 2Cov(X, Y)$$

Case Study: Smart City Traffic + Weather System

Design an intelligent system predicting **traffic congestion** using **transportation and climate variables**.

Step 1: Define Random Vector

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix}$$

Where:

- X_1: Traffic flow (vehicles/min)
- X_2: Average vehicle speed (km/h)
- X_3: Rainfall intensity (mm/hr)
- X_4: Temperature (°C)
- X_5: Air pollution index (AQI)

Step 2: Mean Vector

Example values:

- Avg traffic flow = 120 vehicles/min
- Avg speed = 35 km/h
- Avg rainfall = 2 mm/hr
- Avg temperature = 32°C
- Avg AQI = 180

$$\mu = E[X]$$

Step 3: Covariance Matrix

Interpretation examples:

- $Cov(X_1, X_3) > 0$
  More rain → more congestion.
- $Cov(X_2, X_3) < 0$
  More rain → lower speed.
- $Cov(X_1, X_5) > 0$
  More traffic → higher pollution.
- $Cov(X_4, X_5) < 0$
  Higher temperature → pollution disperses.

Step 4: Joint Gaussian Model

Assume

$$X \sim N(\mu, \Sigma)$$

Reason:

- Aggregation of many small effects (Central Limit Theorem)
- Sensor noise often Gaussian

- Easier analytics (closed-form solutions)

Thus the system is defined by:

- **Mean** → typical conditions
- **Covariance** → interdependencies

Step 6: Transformation

Define congestion index

$$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$$

This is a **linear transformation**

$$Y = a^T X$$

Inference

If

$$X \sim N(\mu, \Sigma)$$

then

- Y is also **Gaussian**

Mean:

$$E[Y] = a^T \mu$$

Variance:

$$Var(Y) = a^T \Sigma a$$

Key Takeaways

- Random vectors represent multiple random variables compactly.
- Mean vector and covariance matrix characterize their statistical behavior.
- Linear transformations modify mean and covariance as:

$$\mu_Y = A\mu_X + b$$

$$C_{YY} = AC_{XX}A^T$$

- Multivariate Gaussian distributions extend the Gaussian concept to N-dimensions.
- Uncorrelated Gaussian variables lead to factorization of the joint PDF.

If you'd like, I can also ↪ create a **1–2 page "exam cheat sheet" version of Lecture 20** (theorems, formulas, and derivations most likely to appear in exams).

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Cse400 Lecture20 Scribe

CSE 400: Fundamentals of Probability in Computing

Lecture 20: Linear Transformations and Expectations of Multiple Random Variables

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Outline

- Statistical Representation of Random Vectors
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- Mapping properties under Y = AX + b
- Multivariate Gaussian Distributions
- From 1D/2D cases to N-dimensional Joint PDFs
- Special Cases and Properties
- Implications of zero correlation and matrix factorization

Random Vector Representation

Vector Form of Multiple Random Variables

For random variables:

X1, X2, ..., XN

We represent them as a vector:

X = [X1, X2, ..., XN]^T

Definition

X is called a **random vector**.

Properties

- Provides a compact representation of multiple random variables.

Mean Vector

Expectation of a Random Vector

Mean Vector Definition

The mean vector of X is:

$\mu X = E[X]$

Interpretation

- Represents the average value of the random vector.
- Each element corresponds to the mean of one random variable.

Structure

$\mu X = [E[X1], E[X2], ..., E[XN]]^T$

Correlation Matrix

Correlation Between Multiple Random Variables

Definition

$RXX = E[XX^T]$

Where:

$RXX(i,j) = E[Xi\ Xj]$

Matrix Representation

RXX =

[E[X1X1] E[X1X2] ... E[X1XN]] [E[X2X1] E[X2X2] ... E[X2XN]] [... ... ...] [E[XNX1] E[XNX2] ... E[XNXN]]

Property

RXX is a **symmetric matrix**.

Covariance Matrix

Definition

$CXX = E[(X - \mu X)(X - \mu X)^T]$

Example Form

CXX =

[Var(X1) Cov(X1,X2)] [Cov(X2,X1) Var(X2)]

Property

Cov(Xi,Xj) = Cov(Xj,Xi)

Thus the matrix is symmetric.

Linear Transformation of Random Vectors

Model

Consider the transformation:

Y = AX + b

Where:

- A : Transformation matrix
- b : Constant vector
- X : Input random vector

Interpretation

A scales and rotates the data. b shifts the center.

Element-wise Representation

Y1 = a11X1 + a12X2 + ... + a1NXN + b1

Y2 = a21X1 + a22X2 + ... + a2NXN + b2

...

YN = aN1X1 + aN2X2 + ... + aNNXN + bN

Mean of Transformed Vector

For Y = AX + b:

μY = E[Y]

Derivation

Substitute Y:

μY = E[AX + b]

Apply linearity:

$\mu Y = E[AX] + E[b]$

Since b is constant:

$E[b] = b$

Thus:

$\mu Y = A\mu X + b$

Correlation Matrix Transformation

Definition:

$RYY = E[YY^T]$

Substitute $Y = AX + b$:

$RYY = E[(AX + b)(AX + b)^T]$

Expansion

$RYY = E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$

Applying Expectation

$RYY = AE[XX^T]A^T + AE[X]b^T + bE[X]^TA^T + bb^T$

Final Result

$RYY = ARXXA^T + A\mu Xb^T + b\mu X^TA^T + bb^T$

Covariance Matrix Transformation

Key observation:

$Y - \mu Y = (AX + b) - (A\mu X + b)$

$Y - \mu Y = A(X - \mu X)$

Derivation

$CYY = E[(Y - \mu Y)(Y - \mu Y)^T]$

Substitute:

$CYY = E[A(X - \mu X)(X - \mu X)^TA^T]$

Result

$CYY = ACXXA^T$

Observation

Shift b does **not affect covariance**.

Summary of Linear Transformation

If $Y = AX + b$:

Mean:

$\mu Y = A\mu X + b$

Covariance:

$CYY = ACXXA^T$

Correlation:

$RYY = ARXXA^T + A\mu Xb^T + b\mu X^TA^T + bb^T$

Numerical Example

Given transformation:

$Y = AX$

Compute:

- Mean vector of Y
- Correlation matrix of Y
- Covariance matrix of Y

Solution: Mean

$\mu Y = A\mu X$

Example calculation:

$\mu Y = [1\ 0\ -1;\ 0\ -1\ 1;\ -1\ 1\ 0] * [2\ -1\ 1]^T$

Component calculations:

1st element:

$(1)(2) + (0)(-1) + (-1)(1)$

2nd element:

$(0)(2) + (-1)(-1) + (1)(1)$

3rd element:

$(-1)(2) + (1)(-1) + (0)(1)$

Result:

$\mu Y = [1, 2, -3]^T$

Correlation Matrix Solution

Use:

RYY = ARXXA^T

Step 1: Compute ARXX

Step 2: Multiply result by A^T

Final:

RYY =

[17 −6 −9] [−6 14 −8] [−9 −8 19]

Covariance Matrix Solution

Use relation:

CYY = RYY − μY μY^T

Outer Product

μY μY^T =

[1 2 −3] [2 4 −6] [−3 −6 9]

Subtract

CYY =

[17 −6 −9] [−6 14 −8] [−9 −8 19]

minus

[1 2 −3] [2 4 −6] [−3 −6 9]

Final covariance matrix obtained after subtraction.

Homework Assignment

Given matrices CXX and RXX.

Find mean vector μX.

Relation:

CXX = RXX − μX μX^T

Rearrange:

μX μX^T = RXX − CXX

Diagonal elements reveal components of μX.

Gaussian Distributions

Univariate Gaussian

$fX(x) = (1 / \sqrt{(2\pi\sigma^2)}) \exp(-(x - \mu)^2 / (2\sigma^2))$

Joint Gaussian (2D)

For variables X and Y:

$fXY(x,y) =$

$1 / (2\pi \sigma x \sigma y \sqrt{(1 - \rho^2)})$

$\exp(-1/(2(1-\rho^2)) [(x-\mu x)^2/\sigma x^2 - 2\rho(x-\mu x)(y-\mu y)/(\sigma x \sigma y) + (y-\mu y)^2/\sigma y^2])$

Multivariate Gaussian Distribution

For random vector:

$X = [X1, X2, ..., XN]^T$

General PDF

$fX(x) =$

$1 / \sqrt{((2\pi)^N \det(CXX))}$

$\exp(-1/2 (x - \mu X)^T CXX^{-1} (x - \mu X))$

Components

μX : Mean vector

CXX : Covariance matrix

Uncorrelated Gaussian Variables

If:

$Cov(Xi, Xj) = 0$ for $i \neq j$

Then covariance matrix becomes diagonal.

Result:

Joint PDF factorizes:

$f_X(x) = \Pi f_{X_i}(x_i)$

Thus independent Gaussian components appear.

Homework Proof

Prove:

$Var[X + Y] = Var[X] + Var[Y] + 2Cov(X,Y)$

Case Study: Smart City Traffic + Weather System

Random vector variables:

X_1 : Traffic flow

X_2 : Average speed

X_3 : Rainfall intensity

X_4 : Temperature

X_5 : Air pollution index

Thus:

$X = [X_1, X_2, X_3, X_4, X_5]^T$

Mean Vector Example

Average values example:

Traffic = 120 Speed = 35 Rainfall = 2 Temperature = 32 AQI = 180

$\mu = E[X]$

Covariance Interpretation

Examples:

$Cov(X_1,X_3) > 0$: Rain increases congestion

$Cov(X_2,X_3) < 0$: Rain reduces speed

$Cov(X_1,X_5) > 0$: Traffic increases pollution

$Cov(X_4,X_5) < 0$: Temperature disperses pollution

Joint Gaussian Model

Assume:

$X \sim N(\mu, \Sigma)$

Reasons:

- Aggregation of many small effects
- Sensor noise Gaussian
- Closed form analytics

System defined by:

Mean → typical conditions

Covariance → interdependencies

Linear Transformation Example

Define congestion index:

$Y = 0.6X_1 - 0.4X_2 + 0.8X_3$

This is linear transformation:

$Y = a^T X$

If:

$X \sim N(\mu, \Sigma)$

Then:

Y is Gaussian

Mean:

$E[Y] = a^T \mu$

Variance:

$Var(Y) = a^T \Sigma a$

Key Takeaways

- Random vectors represent multiple variables.
- Mean and covariance describe system behavior.
- Linear transformations modify mean and covariance.
- Multivariate Gaussian models high-dimensional uncertainty.
- Uncorrelated Gaussian variables factorize into independent PDFs.

I’ve added the **entire lecture scribe into the Canvas** so you can now view and edit it as a structured document.

You can:

- ↳ Scroll through the full **Lecture 20 scribe**
- ↳ Edit or annotate it
- ↳ Ask me to **add derivation steps, formatting, or exam highlights**
- ↳ Ask for **LaTeX formatting improvements or diagrams**

If you'd like, I can also:

- ↳ Convert it into a **clean exam-ready PDF format**
- ↳ Add **boxed formulas for quick revision**
- ↳ Turn it into a **short 3–4 page exam cheat sheet**.